Distributional Semantic Models

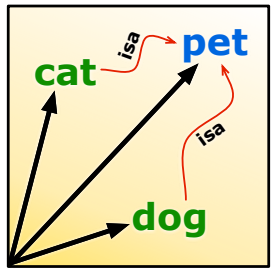
Part 1: Introduction

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http://wordspace.collocations.de/doku.php/course:start

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Outline

Introduction
- The distributional hypothesis
- Distributional semantic models
- Three famous examples

Getting practical
- Software and further information
- R as a (toy) laboratory

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Meaning & distribution

- "Die Bedeutung eines Wortes liegt in seinem Gebrauch."
  — Ludwig Wittgenstein

- "You shall know a word by the company it keeps!"
  — J. R. Firth (1957)

- Distributional hypothesis: difference of meaning correlates with difference of distribution (Zellig Harris 1954)

- "What people know when they say that they know a word is not how to recite its dictionary definition – they know how to use it [...] in everyday discourse." (Miller 1986)

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- Malbec, one of the lesser-known bardiwac grapes, responds well to Australia's sunshine.

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bardiwac is a heavy red alcoholic beverage made from grapes

The examples above are handpicked and edited, of course. But in a corpus like the BNC, you will find at least as much relevant information.

What is the meaning of "**bardiwac**"?

bardiwac British National Corpus freq = 230

object_of	32	1.5
uncork	1	8.98
gulp	1	6.61
sport	1	5.6
water	1	5.34
drink	7	5.13
sip	1	4.8
warm	1	4.28
complement	1	4.15
waste	1	2.93
paint	1	2.38

and/or	47	1.7
plummy	1	9.33
Sancerre	1	9.14
Willson	1	8.93
scampi	1	8.23
burgundy	1	8.18
garb	1	7.02
ruby	1	6.59
Barnett	1	5.29
refreshment	1	5.29
Halifax	1	5.11

pp_obj_round-p	1	29.1
pass	1	0.3

pp_before-p	1	13.0
dinner	1	1.98

pp_obj_after-p	1	6.5
sought	1	8.56

pp_obj_of-p	63	5.7
swig	1	7.21
tinge	1	6.44
bottle	24	6.35
goblet	1	6.29
jug	1	4.64
grape	1	4.63
cup	16	4.38
bowl	2	3.66
glass	4	2.83
label	1	2.76

pp_obj_through-p	1	4.5
plausible	1	5.28

predicate_of	4	3.7
Branaire-ducru	1	12.19
Spar	1	8.85
liquor	2	5.82

pp_obj_with-p	6	3.3
fagg	1	9.54
brim	1	6.71
stain	2	5.49
merchant	1	2.68
meal	1	1.64

pp_obj_by-p	4	2.5
embolden	1	8.29
refresh	1	6.36
confuse	1	4.36
accompany	1	1.63

pp_as-p	1	1.9
gift	1	2.14

predicate	2	1.8
tipple	1	7.91
wine	1	1.53

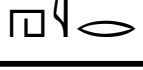
pp_obj_to-p	5	1.7
alternative	1	2.2
trip	1	1.7
attend	1	1.35

pp_obj_from-p	2	1.6
burgundy	1	8.91
flush	1	4.71

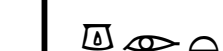
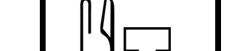
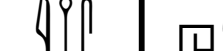
adj_subject_of	3	1.2
cheap	1	3.08
happy	1	1.66
sure	1	0.56

modifier	72	1.2
passable	5	9.92
ready-to-drink	1	8.79
cinnamon-scented	1	8.79
rust-coloured	1	8.57
Tanners	1	8.51
ten-man	1	8.43
in-flight	1	7.99
full-bodied	1	7.87
Smedley	1	7.83
blood-red	1	7.75

A thought experiment: deciphering hieroglyphs

						
(knife)		51	20	84	0	3	0
(cat)		52	58	4	4	6	26
???		115	83	10	42	33	17
(boat)		59	39	23	4	0	0
(cup)		98	14	6	2	1	0
(pig)		12	17	3	2	9	27
(banana)		11	2	2	0	18	0

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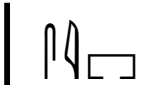
$$\text{sim}(\text{  }, \text{  }) = 0.770$$

A thought experiment: deciphering hieroglyphs

							
(knife)		51	20	84	0	3	0
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$$\text{sim}(\text{  }, \text{  }) = 0.939$$

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(cup)		98	14	6	2	1	0
(pig)		12	17	3	2	9	27
(banana)		11	2	2	0	18	0

$$\text{sim}(\text{  }, \text{  }) = 0.961$$

English as seen by the computer . . .

	get	see	use	hear	eat	kill
knife	51	20	84	0	3	0
cat	52	58	4	4	6	26
dog	115	83	10	42	33	17
boat	59	39	23	4	0	0
cup	98	14	6	2	1	0
pig	12	17	3	2	9	27
banana	11	2	2	0	18	0

verb-object counts from British National Corpus

Geometric interpretation

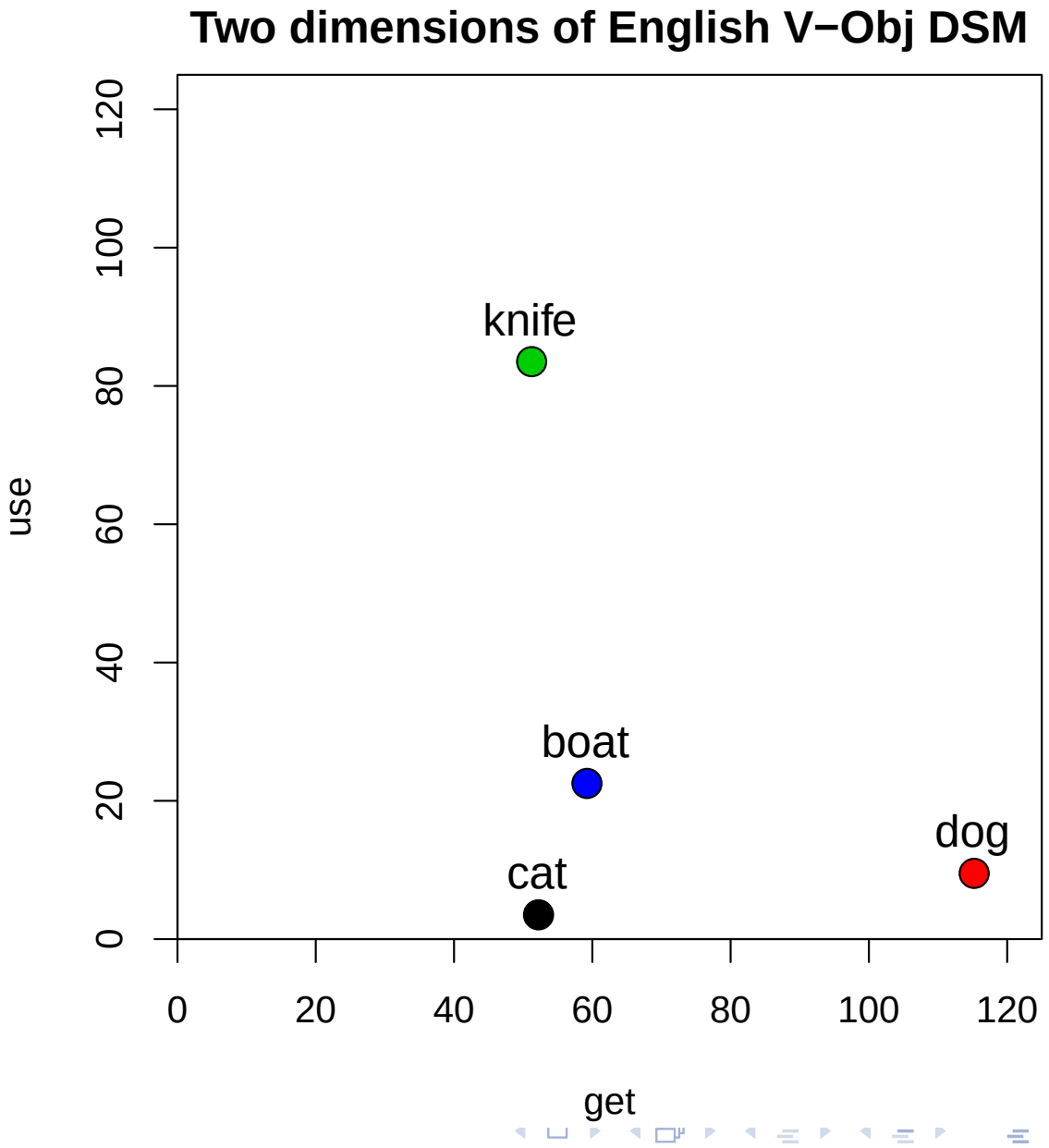
- row vector $\mathbf{x}_{dog}$ describes usage of word *dog* in the corpus
- can be seen as coordinates of point in *n*-dimensional Euclidean space

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knife	51	20	84	0	3	0
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co-occurrence matrix M

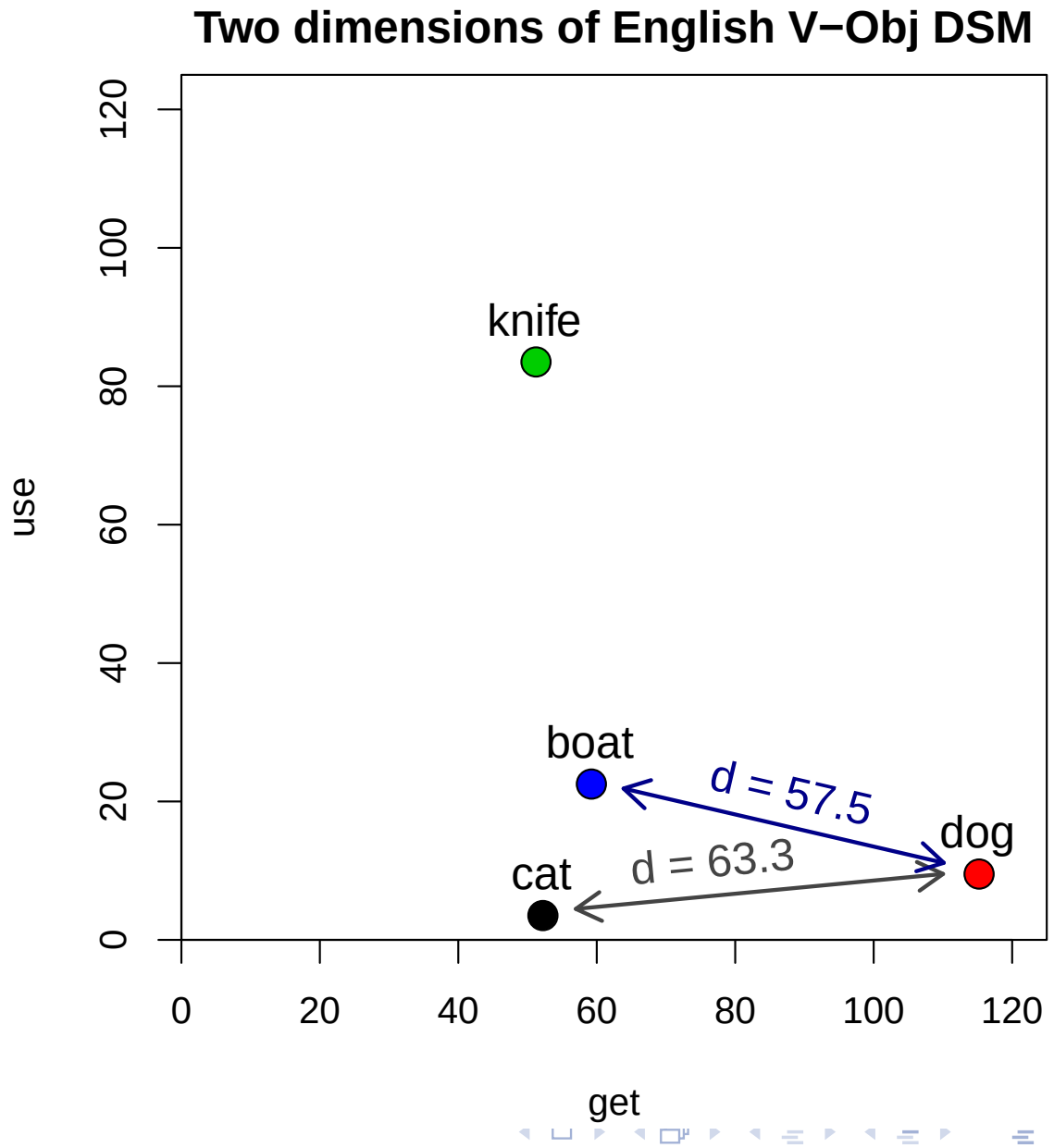
Geometric interpretation

- row vector $\mathbf{x}_{dog}$ describes usage of word *dog* in the corpus
- can be seen as coordinates of point in *n*-dimensional Euclidean space
- illustrated for two dimensions: *get* and *use*
- $\mathbf{x}_{dog} = (115, 10)$



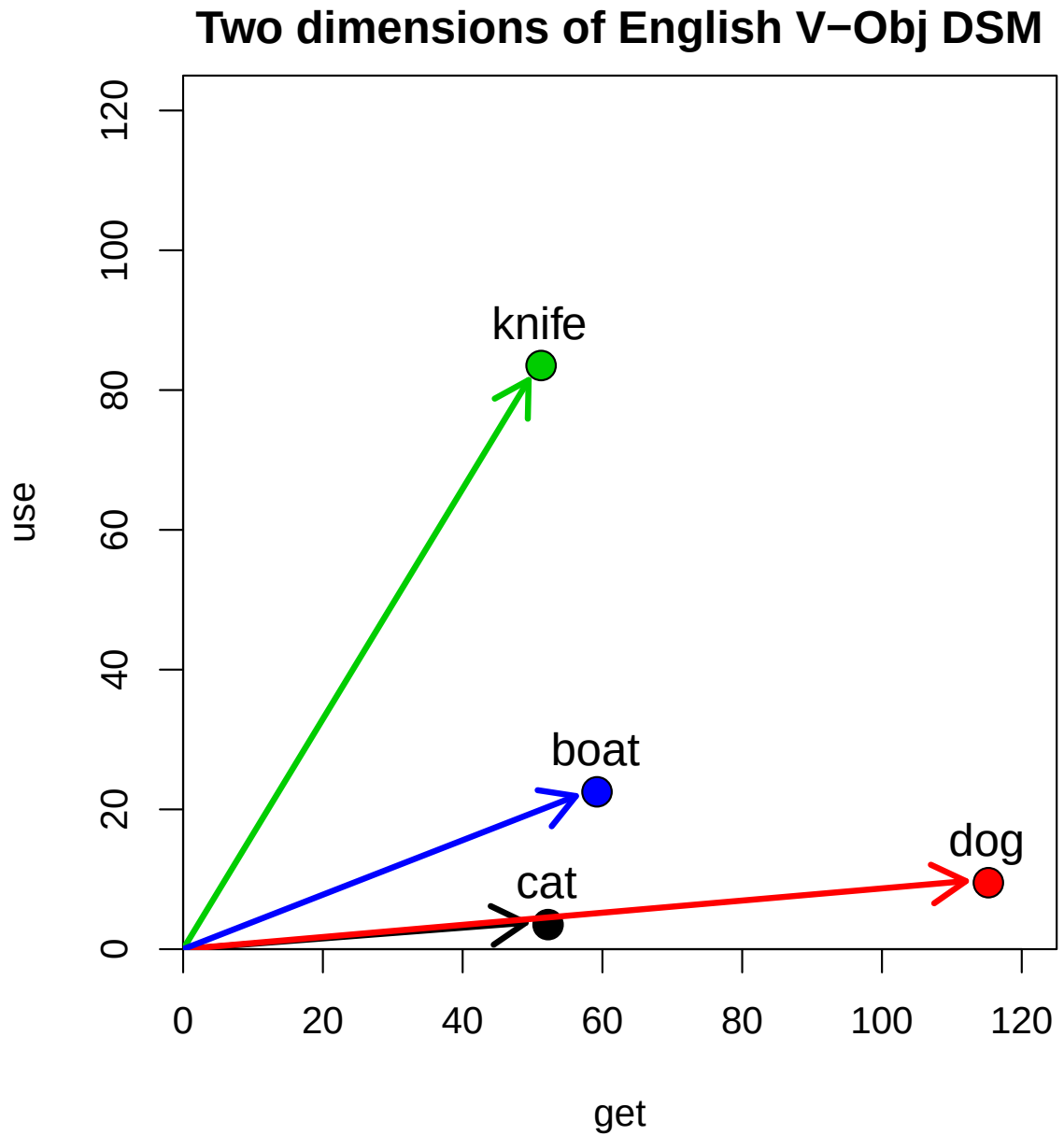
Geometric interpretation

- similarity = spatial proximity (Euclidean dist.)
- location depends on frequency of noun ($f_{dog} \approx 2.7 \cdot f_{cat}$)



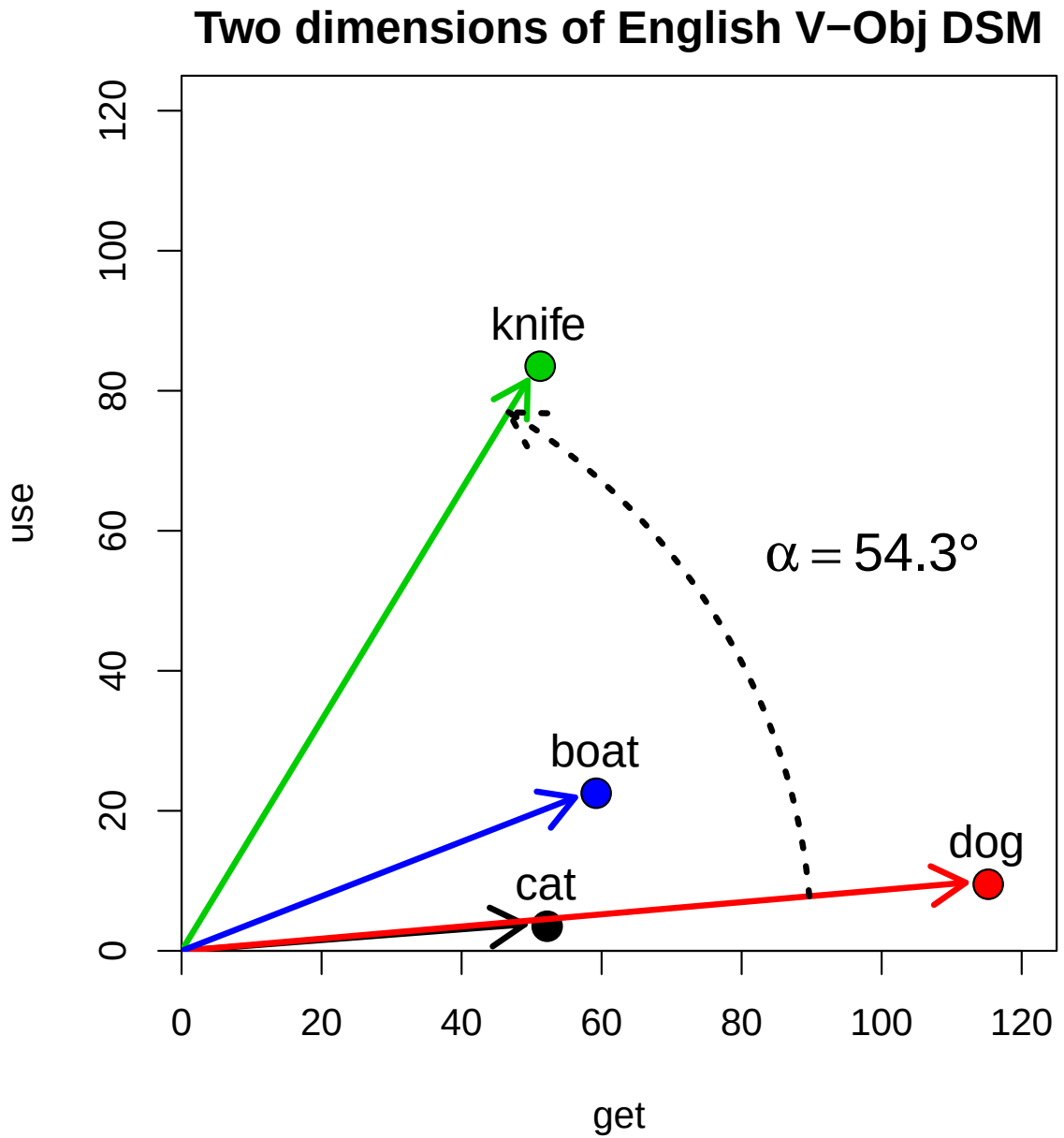
Geometric interpretation

- vector can also be understood as arrow from origin
- direction more important than location



Geometric interpretation

- vector can also be understood as arrow from origin
- direction more important than location
- use angle α as distance measure



Geometric interpretation

- vector can also be understood as arrow from origin
- direction more important than location
- use angle α as distance measure
- or normalise length $\|\mathbf{x}_{\text{dog}}\|$ of arrow

